

Monu Kumar Pathak

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Qualifications	University	Programme	Year
M.Tech Specialization:	IIT Bombay Optimization and Data Science	IEOR	2023–25

WORK EXPERIENCE

Kinaxis India | Software Developer - I (Algorithms) June 2025 - Present

Reprogramming LLMs for Time Series Forecasting

- Implemented TIME-LLM on frozen **LLaMA-7B** by training only **0.2% params** using patch reprogramming
- Built **RevIN normalization** pipeline to handle distribution shift & improve **few-shot** & zero-shot performance
- Designed **Prompt-as-Prefix** with statistical context (**trend, lags, min/max**) for precise numerical grounding
- Achieved SOTA in **36/40** benchmarks with **12%** lower **MSE** vs **GPT4TS** and **20%** lower **MSE** vs **TimesNet**
- Improved zero-shot forecasting **MSE** by **22%** while reducing trainable params by **71%** vs **QLoRA** fine-tuning

Statistical Convolution Modeling for Distribution Estimation

- Developed a Statistical model to predict the distribution formed by convolution of two probability distributions
- Utilized **KS** test, **AIC** & **BIC** to evaluate goodness of fit, & **PP** & **QQ** plots for visual validation of distribution

Research Work

RL for Dynamic Job Scheduling | Master's Thesis

Jan 2025 - June 2025

- Developed an **RL**-based agent to optimize job scheduling across multiple machines based on its system state
- Implemented a **DQN** agent from scratch & developed a SimPy simulation for production plant data generation
- Trained DRL agent using a custom reward function to predict optimal reward for a given state & action pair
- Achieved a **42%** improvement over standalone strategies & **20%** gain compared to existing OR & heuristics

X.A.I. for Advance Simulation Data Analysis | Master's Thesis

June 2024 - Jan 2025

- Implemented **Data Farming** techniques, integrating the **explainable AI** to analyze large simulation data
- Performed **1M+** simulations & clustered simulation output data using **K- Means** & **DBSCAN** algorithms
- Extracted insights & improved transparency with explainability in **Neural Networks** and **Random Forest**
- Utilized **SHAP** to calculate **marginal contribution** of each input feature on model output for explainability

Internship

Cognitio Analytics LLC | Data Science Research Intern

May 2024 - Jul 2024

- Reviewed over **10 research papers** on Generative AI and documented their use case into the medical domain
- Preprocessed EHR data containing **58,000** hospital admissions, **38,000+** adult patients and their diagnosis
- Developed **LSTM** & **Transformers** based models to predict patient next visit embedding from medical history

PROJECTS

Commonsense-Enabled LLMs

Master's Seminar

- Fine-tuned **GPT-1** model to accurately generate, & enhance contextual commonsense **reasoning** & inferences
- Leveraged the **ConceptNet** and **ATOMIC** knowledge graphs with over **40M** and **877K** tokens for training
- Achieved **perplexity** of **11.14** & **BLEU-2** score of **15.10**, reflecting quality & novelty in sentence generation

Privacy Robust Recommendation Engine

[Project Link](#)

- Goal: Development of an **Open-Source** Privacy-Robust Recommendation System using Differential Privacy
- Applied **3 privacy-enhancing** algorithms by injecting **Laplace noise** at input, processing & output stages
- Implemented **Matrix Factorization** from scratch, including **Alternating Least Squares(ALS)** & **SGD**
- Achieved **RMSE** of **0.92** on perturbed data with 3 as privacy parameter, compared to **0.85** on clean data

COURSES AND CERTIFICATIONS

- Machine Learning Techniques
- Mathematical Optimization Techniques
- Deep Learning for NLP
- Engineering Statistics
- Modelling & Computation
- Simulation & Data Analysis

SKILLS

- Languages:** Python, Java, SQL, Kotlin
- Softwares & Tools:** Hugging Face, Github, Azure ML
- Libraries:** Scikit-Learn, Pytorch, NLTK, Pandas, Matplotlib, Tensorflow, Numpy, Scipy, Optuna, Shap